Chapter 4

Cardiopulmonary bypass

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Complications of bypass are discussed together with off-pump surgery on 🕮 p305. ECMO is described on 🕮 p500. Ventricular assist devices are discussed on 🕮 p496. Myocardial ischemia-reperfusion injury, stunning, and hibernation are discussed on 🕮 pp74–75. Spinal cord protection is discussed in more detail on 🕮 p436.

Cardiopulmonary bypass circuit 1

Key facts

Cardiopulmonary bypass (CPB) provides a still, bloodless heart while circulation to the rest of the body is maintained. Alternatives to standard cardiopulmonary bypass, including circulatory arrest and off-pump surgery, are discussed on 📖 p121. There are three essential functions of CPB:

- Oxygenation.
- Ventilation.
- Circulation.

The other important function of CPB is temperature control.

The CPB circuit

Desaturated blood drains from the RA or vena cava via venous cannulas and the venous line (1) to a reservoir (3). A pump (4) propels blood from the venous reservoir through a membrane oxygenator (5), followed by an arterial filter (6), into the patient's aorta via the arterial line and the aortic cannula (7). Adjuncts to the basic circuit include:

- Venous occlusion (an adjustable clamp on the venous line).
- Ports for drug and fluid infusions to the venous reservoir.
- Cardioplegia delivery system using a separate roller pump.
- Source of oxygen, air, CO_2 and anesthetic gases.
- Sampling ports, and in-line blood gas and temperature monitors.
- Bypass line around the arterial filter in case the filter obstructs.
- Low-level reservoir alarm.
- Ultrafiltration.
- Suction tubing leading to cardiotomy reservoir via a pump for removing blood from heart (vents) or the surgical field (pump suckers).

The standard components of the CPB circuit

- Atrial cannula (right atrial, caval, femoral and/or axillary).
- Venous line (PVC ½ inch diameter 12mm).
- Venous reservoir (integrated with oxygenator).
- Venous outlet (PVC ⅜ inch 8mm).
- Pump (peristaltic roller pump or centrifugal).
- Oxygenator (membrane oxygenator, defoamer and heat exchanger).
- Arterial filters macro and micro (300 micrometer) and bubble detector.
- Arterial line (PVC ⅜ inch).
- Arterial cannula (aortic, femoral, or axillary).

Venous cannulas

The technique of venous cannulation is described on 📖 p110. There are four main types of venous cannulas: Ross basket, two-stage, bicaval (single stage), and peripheral venous cannulas. The Ross basket is a large thimble-shaped metal tip, perforated with several large holes, designed to sit in the RA, so that the rim of metal is flush with the atrial purse-string. The two-stage venous cannula is tapered and reinforced with wire to prevent

kinking. It is placed in the RA (usually so that an external marker is flush with the atrial purse-string) ensuring its perforated tip in the IVC, and the proximal perforation sits close to the SVC. Bicaval cannulas can be placed in each cava either directly (metal tipped) or via the right atrium (PVC). **Total CPB** completely eliminates venous return to the RA by placing snares around the caval cannulas. Two-stage cannulation is quick, and provides good venous drainage of the heart in a range of positions.

- Total CPB (bicaval cannulation with IVC and SVC snared) provides better myocardial cooling, but can result in poor SVC or IVC drainage if either cannula malpositioned or blocked, causing venous congestion.
- Total CPB prevents myocardial rewarming as a result of systemic venous return more effectively than other venous cannulation.
- Total CPB prevents airlocks (📖 p118) when opening the right side of the heart, although if using vacuum-assisted venous drainage (VAVD) airlocks occur only very rarely so snares are not necessary.
- Direct bicaval cannulation allows best exposure of the mitral valve.
- Ross basket avoids the risk of damaging the IVC during cannulation.
- For most surgery (coronary, aortic root, ascending, and arch work) a two-stage cannula provides good drainage and access.
- Various combinations of open or percutaneous femoral, internal jugular, and axillary venous cannulas are useful in minimally invasive (📖 p346) and reoperative (📖 p344).

Venous drainage

Venous drainage is by: (1) gravity siphonage: *the venous reservoir must be below the level of the patient, and the line must be full of fluid*, or (2) vacuum assist (VAVD). Inadequate venous drainage leads to the heart distending with blood, increasing myocardial oxygen demand which can lead to ischemia (📖 p113). Venous drainage is increased by ensuring the cannulas are in the correct position, and by increasing:

- The amount of vacuum assist (maximum of –60mmHg).
- CVP which is dependent on intravascular volume and compliance.
- Difference in height between the patient and the venous reservoir.
- Lumen of venous cannulas (36F for two-stage, up to 34F for bicaval).

Problems with venous drainage

- Inadequate venous drainage is caused by:
  - Malposition of the venous cannula: the CVP should be <2–6mmHg. If it is higher always check venous cannula for problems (📖 p113) as cerebral edema can result from inadequate SVC drainage.
  - Positioning the heart, kinking, or clamping the lines
  - Hypovolemia and vasodilation.
  - Airlock interrupting siphon drainage, loss of vacuum assist (📖 p118)
  - Tearing the IVC during insertion of a two-stage cannula (📖 p118).
- 'Chatter': venous drainage may be faster than blood return, leading to intermittent collapsing of the cava and atrium against the cannula, stopping drainage, rapidly alternating with atria filling with blood, causing juddering. To correct this, the perfusionist reduces venous drainage by increasing the amount of clamp or occlusion of the venous line.
- Hemolysis results from excessive VAVD and air admixture.

Cardiopulmonary bypass circuit 2

Arterial cannulas

The key is that the narrowest section of the bypass circuit is the part of the arterial cannula that enters the aorta; this limits flows and it should therefore be as short as possible. High flows through narrow cannulas generate high-pressure jets which may dislodge atheroma, and pressure gradient >100mmHg causing hemolysis, turbulence, and cavitation. Most standard 24F adult arterial cannulas provide adequate flows at gradients <100mmHg in patients of any size. A variety of options is available:

- Diffusion tips (several side perforations rather than one central one) are designed to reduce high-pressure jets. The Sarns Soft-Flow® is an example (vs. the Sarns bevelled tip).
- Vents to allow additional de-airing once connected.
- Flanges that may or may not be removable to improve hemostasis.
- Flexible lumen so cannula can be positioned out of the field.
- Seldinger technique for very small incisions (Edwards Fem-Flex®).
- Removable intra-aortic filter to capture debris (Edwards EMBOL-X®).

Tubing and connectors

Tubing and connectors are biocompatible, i.e., non-toxic, non-immunogenic, non-allergenic, and non-mutagenic. They should be waterproof, transparent, flexible but strong, inert, smooth, and easy to manufacture in bulk. Tubing is usually PVC and may be heparin bonded (which reduces complement activation). Traditionally *venous tubing diameter is ½ inch (12mm), and arterial tubing is $^{3}/_{8}$ inch (9mm)*: the large tubing size reduces resistance to blood flow, and the difference also minimizes the risk of incorrect connections.

Pumps

Pumps are either peristaltic or centrifugal. *Peristaltic or roller pumps* consist of a section of tubing inside a metal housing. Two rollers mounted at 180° to each other rotate so that one roller is compressing the tubing at all times. Blood flow is achieved by pushing the blood ahead of the moving roller. Excessive compression causes hemolysis. Inadequate compression also induces hemolysis and reduces forward blood flow which depends on the size of the tubing, the length undergoing compression, and the rpm. Roller pumps are used to provide arterial flow to the patient, for administering cardioplegia and for providing suction (by reversing the direction of tubing in the metal housing). If the line becomes occluded the roller pumps continue to exert pressure (positive or negative). Tubing and connector rupture is avoided by using pressure regulators. High negative pressures are avoided by the perfusionist either asking the surgeon to reposition occluded lines or by reducing the rpm.

Centrifugal pumps consist of a nest of smooth plastic cones located inside plastic housing, which is rotated at 2000–3000rpm generating a pressure differential. Centrifugal pumps are totally non-occlusive and afterload dependent: increase in downstream resistance decreases the rate of the pump and arterial return to the patient. When the pump is connected to the patient's arterial system, but is not rotating, the arterial line must be clamped as blood and potentially air will flow back into the pump from the

patient. These pumps also become deprimed and stop working if >50mL of air is entrained.

Oxygenators

There are two main types of oxygenator: *membrane oxygenators* which are widely used, and *bubble oxygenators* which are no longer widely used. Most oxygenators come as part of an integral unit housing a heat exchanger, venous reservoir, arterial filter, gas inlet, infusion and sampling ports. Membrane oxygenators can either be true membrane oxygenators that completely separate blood and gas phases, with gas exchange occurring by diffusion, as in the lungs; or the membrane can be full of micropores which increase diffusion. Because the surface area required for adequate gas exchange is so much larger in true membranes, microporous membranes are used in clinical practice. There are two designs of microporous membrane oxygenator: the hollow fibre design with blood flow either inside or outside the fibres, and the folded envelope design.

Bubble oxygenators consist of two chambers. Venous blood flows into the first mixing chamber, and gas is pumped into it through a screen causing thousands of tiny bubbles to form: it is these across which gaseous exchange takes place. The mixture then flows into a defoaming chamber where it comes into contact with defoaming agents silicone Antifoam A and particulate silica. The main practical differences between membrane and bubble oxygenators are:

- Most of the damaging complement activation and inflammatory response occurs in the first few minutes of use of a membrane oxygenator, reducing as blood proteins coat the membrane (compared to bubble oxygenators where the blood–gas interface changes with every new bubble): membrane oxygenators are therefore preferred for long bypass procedures.
- There is a big pressure drop over the membrane oxygenator and the arterial pump is therefore placed after it in the circuit, as opposed to before the bubble oxygenator where the pressure drop is small.
- Ventilation and oxygenation are relatively independent in membrane oxygenators, and it is easier for the perfusionist and anesthesiologist to achieve set parameters by adjusting FiO_2.

Heat exchangers

Heat exchangers can cool or warm blood, controlling the temperature of the patient and of cardioplegia if required. Heat exchangers are an integral part of most oxygenators. They are located proximal to the oxygenator to avoid releasing gas bubbles into the blood, which can occur when the saturated blood is warmed. Water circulates within the heat exchanger between 2–42°C: temperatures outside these ranges risk denaturing proteins. As a result blood can be cooled (at 0.5–1.7°C per minute) more quickly than it can be rewarmed (0.2–0.5°C per minute) as greater temperature differentials can be maintained while cooling. Cooling is slower at lower temperatures for the same reason. The temperature of the patient lags behind the temperature of the perfusate, and the difference between the two is maintained at <15°C.

Pathophysiology of bypass

The pathophysiological changes associated with bypass are due to more than activation of the whole-body inflammatory response as a result of the passage of blood through the non-endothelial circuitry. Changes in temperature, acid–base balance, hemodilution, non-pulsatile flow, drugs, circulating volume, and the mechanics of bypass all contribute to dysfunction of blood constituent cascades and whole organ systems.

Activation of blood constituents

Plasma protein systems

CPB activates five plasma protein systems: the contact system, the intrinsic and extrinsic coagulation pathways, and the complement and fibrinolytic cascades. When blood encounters the non-endothelialized surfaces of the CPB circuit and the operative field, plasma proteins are instantly adsorbed onto the surface to produce a protein layer that varies with the material and the duration of exposure. A dynamic between circulating and adsorbed proteins is established. Heparin-coated circuits change the reactivity of adsorbed proteins but do not reduce thrombogenicity.

- *The contact system* (factors XII, XI, prekallikrein, and kininogen) is activated by non-endothelial surfaces, which in turn activates the intrinsic coagulation pathway, the complement pathway and neutrophils.
- Activated factor XI initiates the *intrinsic coagulation pathway* (📖 p71) which eventually converts prothrombin to thrombin, and fibrinogen to fibrin, providing a major stimulus to coagulation during bypass.
- Tissue factor produced by the damaged tissue surfaces of the wound triggers the e*xtrinsic coagulation pathway* by activating factor VII.
- Factor XIIa produced by the contact pathway activates C1 of the classic *complement pathway*, augmented by the alternative complement pathway which predominates during CPB, to produce anaphylotoxins that increase capillary permeability; alter vasomotor tone; impair cardiac function; activate neutrophils, mast cells, and platelets; and mediate cell and platelet lysis.
- Thrombin produced by activation of the coagulation cascade stimulates endothelial cells to produce tissue plasminogen activator, which cleaves plasminogen to plasmin resulting in progressive *fibrinolysis*.

Cellular systems

CPB activates five cellular systems which mediate systemic inflammation.

- *Platelet activation* via the glycoprotein IIb/IIIa receptor complex by heparin, thrombin, complement, plasmin, platelet activating factor, and hypothermia results in a decrease in platelet numbers and function, and is a major cause of postoperative bleeding.
- Although dilution decreases neutrophil counts during CPB, activation by complement, kallikrein, factor XIIa, and interleukin 1B results in *neutrophils* accumulating in the lungs where they mediate increased capillary permeability and interstitial edema.
- *Monocytes* are activated more slowly than other cellular elements by complement, endotoxins and contact with the CPB circuitry, producing tissue factor and cytokines, and conjugating with platelets.

- *Lymphocytes* including B-lymphocytes, natural killer cells, and T-helper cells are collectively reduced by CPB, reducing γ-interferon production, phagocytosis, and the response to infection for days.
- *Endothelial cells* produce tPA in response to circulating thrombin, activating the fibrinolytic pathway.

Systems dysfunction

Hemostatic

Bleeding and thrombotic complications associated with CPB are related to activation of platelets and plasma proteins, protamine and heparin. Bleeding times after CPB and full reversal of heparinization do not return to normal for 4–12h after bypass. Disseminated intravascular coagulation (DIC) and heparin-induced thrombocytopenia (HIT) and thrombosis (HITT) are discussed on 📖 pp288–289.

Fluid balance

Massive fluid shifts, largely into the interstitium, result from increases in systemic venous pressure, volume loading, reduction in plasma protein concentration as a result of dilution and adsorption onto the CPB circuit, and the inflammatory increase in capillary permeability described.

Endocrine

The combined stressors of surgery, hypothermia, CPB, and non-pulsatile flow trigger a hormonal stress response. Levels of cortisol, epinephrine and norepinephrine rise during bypass and remain raised for at least 24h afterwards, as does blood glucose. Circulating T3 falls below normal range. Release of numerous vasoactive substances, in addition to those described, that act throughout organ systems mean that CPB turns homeostasis into 'physiologic and biochemical chaos'.

Heart

This is described on 📖 p136.

Lung

Pulmonary edema is caused by activation of complement and sequestration of neutrophils in the pulmonary vasculature where they mediate increase in capillary permeability which is compounded by the fluid shifts described. CPB reduces the effect of natural surfactant, compounding the effects of general anesthetic and median sternotomy that predispose to pulmonary dysfunction (📖 p136). CPB increases shunts, reduces compliance and functional residual volume, and can cause ARDS (📖 p163).

Central nervous system

Stroke is mostly due to emboli from cannulating and clamping the aorta.

Kidney

Hemodilution, microemboli, catecholamines, low perfusion pressure, diuretics, hypothermia, and hemolysis impair renal function.

Gastrointestinal

Peptic ulceration is a response to stress, not to CPB per se. Pancreatitis and mild jaundice are not uncommon. Greater permeability of gut mucosa leads to endotoxin translocation, adding to the inflammatory response.

Technique of aortic cannulation

- There are many techniques of cannulation. The one described is a commonly used, safe way of cannulating the heart without an assistant.

Basic cannulation sequence

- Give heparin before dividing the LIMA, or after completing the median sternotomy if no IMA, and secure the CPB lines on the field.
- Hitch the pericardium and place side-towels.
- Palpate aorta and use epiaortic US to assess aortic atheroma.
- Place the aortic purse-strings.
- Check with perfusionist, clamp and divide the CPB lines (📖 p112).
- Check that the systolic pressure is <100mmHg.
- Place a pump sucker to the left of the PA if the ACT is >300.
- Warn the anesthesiologist that you are cannulating.
- Cannulate the aorta and connect the arterial line.
- Check that there is a good swing, and that the line pressure is okay.
- Place the right atrial purse string and cannulate the atria.
- Connect the venous line.
- If the ACT >400 you are now ready to go on bypass (📖 p112).

Aortic cannulation

- Palpate the aorta and use epiaortic ultrasound to identify atheromatous plaques. Pick a soft, clean spot, on the inner curve of the intrapericardial aorta, which is thought to be more resistant than the greater curve and the extrapericardial aorta to dissection and tears.
- Make two diamond-shaped purse-strings with a greater transverse diameter, about 1cm^2, using double-ended 4/0 Ethibond. The easiest way of doing this is two forehand partial-thickness stitches followed by two backhand partial thickness stitches, then the second purse-string, just outside the first, is four forehands. The trick to ensuring that the needle does not go full thickness is to mount it so that it is effectively flat and drive it parallel to the surface of the aorta.
- Place a snare on the sutures and place a clamp on the sutures.
- Now clamp and divide the lines (📖 p112), and check the arterial line for air.
- Take a Debakey's in your left hand and grasp the adventitia caudal to the planned arteriotomy. With an 15-blade in your right hand, and after clearing the adventitia from the cannulation site, scratch the aorta transversely until there is a 5mm incision down to the intima. Insert the blade full thickness. Pull the adventia over the aortotomy to maintain control while you pick up the aortic cannula.
- Hold the cannula so that the bevel is pointing towards the arch, and slide it firmly into the aortotomy up to the flange. Keep hold of the cannula with your right hand and tighten the snare with your left (you can't do this one-handed if you put the mosquito on the snare).
- Now still holding the snare tight with your left hand put a tubing clamp on the aortic cannula with your right hand positioning the clamp against the horizontal bar of the sternal retractor, propping the aortic

cannula in place so that you can tie it firmly to the snare with a heavy silk tie. Remove the plastic cap occluding the cannula.
- Partially release the tubing clamp to fill a gallipot with blood, completely de-airing the cannula. Replace the clamp on the aortic cannula.
- Remove the tubing clamp from the arterial line, and ask the perfusionist to wind up the level of fluid so that you can join the aortic cannula to the arterial line without any air. You can get rid of the air in the arterial line without the perfusionist winding up by cutting the line flush with the clamp and squeezing hard as you connect it to the cannula, but seeing the arterial fluid 'come up' confirms you have the right line flowing in the right direction.
- The perfusionist should confirm that there is a swing and, after infusing a small amount of prime, that the line pressures are satisfactory.

Problems with aortic cannulation

Poor exposure of aorta (short, high aorta or small, low incision)

Clamping aortic adventia, or passing a tape around the aorta, and retracting towards the feet improves exposure for cannulation. Use a McIndoe scissors to develop the plane between the PA and the ascending aorta, extend it by encircling the aorta with your finger and thumb, and pass a Sem clamp carefully from right to left under the aorta, opening it gently so that you can place a wet tape in the jaws.

Can't get the aortic cannula into the aortotomy

If your aortotomy is big enough, difficulty inserting the cannula is usually due to a flap of adventitia. If the systolic BP is low, the aorta will tend to slide away as you cannulate it. Control bleeding using a forceps or a finger, place a pump sucker to the left of the PA, and slide a closed McIndoe scissors into the aortotomy to confirm free passage. If the McIndoe scissors passes easily up to a 1cm, a 24 gauge cannula will.

Cut the purse-string

Should have used two purse strings. Either your purse sting was too small, or visibility was compromised by hematoma. Take a second suture, then ask the scrub nurse or assistant to hold the cannula in place, pushing down gently so that the flange helps control bleeding from the aortotomy. Sew a second purse-string around the aortic cannula, and snare as usual.

Adventitial hematoma

Caused by bleeding from full thickness purse-strings. Divide adventitia with McIndoe scissors. If bleeding is profuse place an extra purse-string.

Aortic root cardioplegia

Siting the cardioplegia cannula depends on whether it will be used as a root vent, proximal anastomosis site, or incorporated into an aortotomy. As a rule of thumb it needs to be far away enough from the sinotubular junction so that the aortic leaflets are not inadvertently damaged, at the highest point of the aorta if is going to be used as a vent, but far enough away from the aortic cannula that the cross clamp can be applied without difficulty. Stockert cannulas require a partial thickness purse-string with 4/0 proline. Flush the plegia so that there is no air in the circuit before connecting.

Technique of venous cannulation

The main advantage of having the aorta cannulated before you put in your atrial purse-strings is the ability to give rapid and controlled transfusion from the pump if handling the right atrium destabilizes the patient.

Atrial cannulation

Place a full thickness purse-string about 2×2cm (2×4cm if a Ross basket is being used) in the face of the atrial appendage, avoiding the right coronary artery in the AV groove, so that the free ends are at 12 o'clock. Snare these and hang them away from you so that their weight pulls the purse-string taut if you grasp the purse-string at 6 o'clock with a Debakey's. Open the purse-string vertically with an 11-blade, widen the incision using a McIndoe scissors, insert the cannula (or Ross basket attached to the end of the venous line) up to the appropriate mark. If you are using a two-stage venous cannula remove the obturator and quickly check that it is in the IVC by palpation. If it does not pass easily into the IVC do not force it. Snare the purse-string and tie it firmly to the venous cannula with a silk tie. Finally connect the venous line to the atrial cannula. There will be some air in the venous line: this does not matter unless you have enough (several feet) to cause an airlock.

Bicaval cannulation

Indirect

Site the first purse-string in the manner described. Site the other one 1–2cm away from the IVC (higher if you plan to 'cross' the caval cannulas). You need to gently retract the right atrium with your left hand, while placing a 2cm^2 triangle sized purse-string where the RA wall seems thick enough to hold it. The heart will decompensate as filling is impaired, so be prepared to release the right atrium in between stitches so that the heart can recover. Pass one cannula into the SVC and the other into the IVC—this can be done through either purse-string. 'Crossing' the cannulas is where the cannulas are placed via the purse-strings furthest away from the respective cavae, which can help exposure if a mitral retractor is not used. Ensure that the cannula in the SVC is rotated so that the bevel faces left, otherwise drainage of the left jugular and subclavian vein will be inadequate. Palpate the SVC and IVC to check that the cannulas are no further than 1cm into the cavae, snare and tie.

Direct

Dissect the pericardium off the SVC, being careful not to injure the phrenic nerve lying on the pleural side, and place a long narrow diamond purse-string on the SVC, well away from the sinoatrial junction. If exposure is poor ask the anesthesiologist to hold ventilation, retract the aorta using the tape or a nerve root retractor in your assistant's right hand, and have your assistant hold the SVC adventia with a forceps in their left hand. Place a pump sucker to the right of the SVC. Hold the SVC opposite your assistant with a forceps and make a 1cm longitudinal incision with an 11-blade. You control bleeding by pushing your forceps against the assistant's, while taking the cannula (which you should bend into an 'L' shape) and inserting

the tip into the SVC. Tie it in place with two heavy silks to stop it spinning out of position. The IVC cannula is placed in the same way as for indirect cannulation. Connect the caval cannulas with a 'Y' connector. Some surgeons clamp the SVC cannula initially while going on bypass via the IVC to avoid a risk of entraining air, but it is just as easy to exclude air from the circuit using venous pressure before you connect it.

Taping the cavae

It is usually quite easy to pass a Sem round the IVC and SVC in order to tape them. Make it even easier by partially dissecting away the pericardial reflection superiorly from both vessels. This will also improve exposure of the mitral valve as the atria can be lifted further away by the retractor.

Retrograde cardioplegia

Site a 1cm^2 triangular purse-string a few cm to the right of the atrial appendage. Bend the cannula 120°, point it away from you, and drop it through the purse-string. It should find its way naturally into the coronary sinus, and you will feel the cannula through the posterior surface of the heart. A manometry line is attached: coronary sinus pressure should be 30–50mmHg while plegia is being given. A drop in pressure implies that the balloon tip has deflated, the catheter is no longer in position or the sinus is perforated. The coronary sinus catheter also can be easily placed under direct vision by opening the RA once on bypass if you are bicavally cannulated.

Problems with venous cannulation

Can't get the venous cannula into the right atrium

If the purse-string and incision are not too small, trabeculae may prevent smooth passage. Put your little finger (middle finger if using a Ross basket) into the atriotomy, then try inserting the cannula again.

Can't get the venous cannula into the IVC or SVC

Either a prominent valve blocks passage of the cannula, or the cavo-atrial junction is distorted by putting the pericardium under too much tension. Try gently probing the IVC with the cannula using your right hand to palpate it and guide it from outside the RA.

Can't get the retrograde cannula into the coronary sinus

Orient the cannula so that it is not twisting around and pointing towards you and away from the coronary sinus once it enters the right atrium: use an external marker such as the position of the plastic handle as a guide. Place your hand under the hearts so you can feel the catheter. If you place a two-stage cannula first it may obstruct easy passage. TEE guidance can be helpful. *Never force it: placing it under direct vision is easier than repairing a perforated coronary sinus.*

Right atrium tears easily

Avoid placing purse-strings too close to the SVC and IVC unless you plan to cannulate them directly. Pick the white, thicker patches of atrial wall. Trying to primarily close tears in dilated, thin-walled atria can rapidly make them significantly worse. Use pledgets. If you make a significant tear, place a second purse-string to prevent entraining air during bypass.

Conduct of bypass

Heparinization

- The ACT should be >300 in order to cannulate.
- The ACT must be >400 in order to commence CPB.
- The ACT must be >480 for hypothermic CPB.
- 3mg/kg (300 units/ kg) is given iv by the anesthesiologist.
- 1mg of protamine is given for every 100U heparin for reversal.
- The ACT is checked after 3min.
- The pump contains 3 units/mL of pump prime (about 5000U).

Prime

Before institution of bypass the tubing is filled with crystalloid (or occasionally blood) to exclude air from the system. This is called priming, and takes ~2L of fluid (usually crystalloid based ± colloid ± 5000U heparin, which causes initial hemodilution, but blood can be used if needed). Large-bore tubing minimizes the gradients needed to achieve adequate flow rates. Minimizing tubing length reduces the amount of prime required. *Retrograde autologous prime* is where the patient's blood is allowed to flow back through the arterial cannula into the tubing, oxygenator, and reservoir displacing up to 1L of prime, with vasoconstrictors to maintain BP if needed. The aim is to reduce hemodilution and transfusion requirements: supporting evidence that this is achieved is mixed.

Dividing the lines

The arterial and venous lines are in continuity when they are handed out by the perfusionist, so that prime can be circulated continuously round the pump. The lines should have been completely de-aired by the perfusionist. Check that there is no residual air, tap connectors to dislodge any bubbles, and check that the perfusionist is ready for you to 'clamp and divide' the lines. Divide them so that tubing length is minimized and they sit flat on the drapes without kinking.

- If you clamp while prime is still being circulated by a roller pump you will cause the pressure behind the clamp to increase until either the perfusionist spots your error or the circuit ruptures.
- If you divide the lines before the perfusionist has clamped their end of the venous line, all the prime in the venous line will be siphoned into the reservoir leaving an empty venous line: refill from the table.

Commencing bypass

Connecting the lines to the cannulas is described on 📖 p109. Try to ensure that that the venous line is uppermost (as this is often mobilized during surgery), that the arterial line is lowest (as this reduces the risk of it being dislodged) and that the cardioplegia and vents are between the two. It is customary not to give heparin until a satisfactory length of conduit is exposed, and not to go on bypass until all conduit has been checked in coronary artery surgery. The ACT >400 cut-off is based on early observational data that no thrombosis occurred in the reservoir at this ACT.

- Check the ACT >400 and that any line clamps are removed.
- Tell the perfusionist to 'Go on bypass.'

- The perfusionist gradually removes venous occlusion at the same time as gradually increasing the perfusion flow up to about 2.2L/min, taking care to balance venous return with arterial inflow.
- At full flow the anesthesiologist should stop lung ventilation.

High line pressures

This is when pressures >300mmHg are generated when the perfusionist attempts to transfuse volume from the pump into the aorta before bypass, or on commencing bypass. The aortic cannula may be abutting plaque, up against the opposite wall, not in the aorta, or have caused an aortic dissection (📖 p118). Stop transfusing immediately, check the line is not clamped or kinked, and evaluate the cannulation and aorta site directly, as well as with TEE.

Poor venous drainage

The perfusionist is receiving inadequate venous return, usually because the venous cannula is malpositioned (📖 p103, or because of low venous pressure. Adjust the venous cannula: the RA and RV should collapse. If they are already flat, then poor venous return is due to low CVP (📖 p103).

Sources of blood returning to the heart on bypass

Blood returns to the RA past the caval cannulas if they are not snared, and via the coronary sinus, great cardiac vein, and Thebesian veins. Blood returns to the RV via the RA and the Thebesian veins which drain the myocardium. Blood returns to the LA via the bronchial veins which empty into the pulmonary veins. This flow may be up to 200mL/min, and even greater in patients with COPD. Blood returns to the LV from the aortic root if there is AI (no clamp) or the aortic clamp is not fully on, from the left atrium and the Thebesian veins. Abnormal sources include persistent LSVC (📖 p121), patent ductus arteriosus, systemic to pulmonary artery (Blalog–Taussig) shunt, ASDs, VSDs, and anomalous systemic venous drainage. This blood is warmer and may cause myocardial rewarming.

Venting

Effective venting prevents left heart distension and reduces myocardial rewarming, both reducing the risk of myocardial ischemia; it also improves visibility and exposure, and helps in de-airing. Vent sites are:

- Aortic root (often as a sidearm on the cardioplegia cannula).
- Right superior pulmonary vein (placed into across mitral valve into LV).
- LV apex.
- Pulmonary trunk (left heart venting less satisfactory, via collaterals).

Cardiopulmonary bypass during operation

Flows are maintained at 2.2–2.4L/min/m^2, and the patient is cooled (or the temperature allowed to drift) down to 32°C for most CABG and valve operations. Lower flows are possible at temperatures below 28°C, as oxygen demand (VO_2) decreases. The peripheral vascular resistance is manipulated with vasodilators (NTG) and vasoconstrictors (metaraminol) so that systemic pressure is 60–80mmHg. Choice of temperature and hypothermia is discussed on 📖 pp124–125.

Weaning from bypass

Criteria for discontinuing bypass

- No condition (e.g., bleeding, residual regurgitation, ischemia) that requires bypass to correct.
- Heart largely free of air.
- Ventilating.
- Satisfactory rhythm, ventricular rate >60, paced if necessary.
- Nasopharyngeal temperature 36.5–37.5°C.
- K^+ 4.0–5.5mmol/L.
- pH 7.30–7.50.
- pO_2 >16kPa (120mmHg).

Standard rewarming

Rewarming occurs more slowly than cooling at 0.2–0.5°C/s (📖 p105). Most surgeons ask the perfusionist to rewarm when they are on their last distal anastomosis (CABG), closing the aortotomy (AVR) or the atriotomy (mitral). As the perfusionist increases flows to speed up rewarming and supply increasing oxygen demand, there is an increase in coronary flow which may impair visibility slightly. Once bypass is discontinued the only means of warming the patient is using a sterile warming blanket. With the chest open there is net heat loss, but overwarming to 38°C is associated with adverse neurological outcomes. Hemostasis and chest closure should be carried out efficiently to reduce further heat loss.

De-airing the heart

This is a critical process, particularly important after valve surgery. De-airing in CABG normally consists of needling the grafts proximal to bulldog clamps, and releasing the clamps once air and no longer be seen in the conduit. There are a variety of ways of de-airing after valve surgery. A sensible way is to de-air the heart chambers sequentially is:

- Use CO_2 insufflation into the field for valve surgery.
- As the aortotomy or atriotomy is closed, stop venting the LA and LV.
- Close the suture line as blood completely fills the chamber.
- Vent the aortic root (venting earlier will just suck air into the heart).
- Release any caval snares and ask the perfusionist to 'fill the heart' (the perfusionist partially clamps the venous line, so that the right side of the heart fills with blood, displacing any air present on the right).
- Ask the anesthesiologist to 'inflate the lungs', which dispels air from the pulmonary veins into the LA, which can be aspirated from the dome of the LA or by loosening the LA suture line temporarily.
- Some surgeons shake the patient to dislodge air from underneath valve leaflets and trabeculae, bringing to it to the surface of the chambers: agitating the heart has a similar purpose.
- If a mechanical mitral valve is being frustrated with a Foley catheter, deflate the balloon, remove the catheter and close the atriotomy.
- Lift the heart gently so that the LV apex can be aspirated.
- If an LV vent has been used, agitate the heart until micro air bubbles stop leaving the ventriculotomy, which should then be oversewn.

- 'Empty out' the heart, stop ventilating, put the patient head-down.
- Reduce flows temporarily as the aortic clamp is released.
- Continue venting the aortic root.
- Defibrillate the heart if it is not beating and pace if necessary.
- Agitate the ventricles again, and ask the perfusionist to gently fill the heart so that it is ejecting, and reassess air using TEE.

Sudden arrhythmias and ST segment changes during this period usually reflect coronary air embolism. Resume full bypass and increase MAP to >80mmHg. Re-ops need more aggressive de-airing as adhesions distort tissue, creating pockets of air particularly in the pulmonary veins which take longer to shift.

Discontinuing bypass

The perfusionist, anesthesiologist, and surgeon should work together to monitor the effect of gradually reducing CPB flow on the heart. The perfusionist gradually increases occlusion on the venous line at the same time as reducing arterial flow. Increasing occlusion fills the heart. Ventricular function can be assessed by the surgeon (the RV should still wrinkle, the heart should not rise much above the pericardial reflection and the PA should feel reasonably soft to palpation), and by the CVP which should be 10–12mmHg. It is possible to come completely off bypass at any time during weaning. If is usually safest to come off with the heart underfilled rather than overfilled, as the increased afterload and preload increases myocardial oxygen demand at a time when myocardial stunning may mean that the heart is poorly equipped to deal with this. Once you are off bypass, if the heart appears to be contracting well you can ask the perfusionist to transfuse some of the remaining volume in the venous reservoir. Some teams give vasodilators so that all of the volume remaining in the venous reservoir can be transfused back to the patient. Poor LV is managed by a period of partial bypass while preload, afterload and contractility is carefully adjusted using inotropes, pressors, and filling. The goal is MAP 75–80 and CVP 6–12mmHg.

- Once bypass is discontinued the venous cannula can be removed.
- The blood in the venous line is siphoned by the perfusionist.
- The venous cannulation site is snared.
- Protamine is given slowly, and hemostasis secured.
- Volume is transfused as appropriate into the aorta.
- The aortic cannula is removed and both cannulation sites oversewn.

Protamine

Giving protamine on bypass leads to catastrophic thrombus formation in the circuit causing systemic emboli and failure of the oxygenator. Protamine is given once bypass has been permanently discontinued, in a slow iv push. 1mg of protamine is given for every 100U of heparin given. Pump suckers should be removed as soon as protamine is started. The ACT is checked after 5min and further protamine given if required. Protamine is usually associated by a transient drop in systemic pressures, and occasionally with one of four types of adverse reaction (📖 p116).

Emergency bypass scenarios 1

Protamine reaction

Protamine is a polycationic protein derived from salmon sperm, which binds ionically with anionic heparin in a 1:1 ratio, to produce a stable precipitate. Protamine contains two active sites, one that neutralizes heparin and one that has a mild anticoagulant effect. There is therefore a theoretical risk that excess protamine is anticoagulant: it is probably only clinically relevant at 3× the appropriate dose. Protamine has a mild direct antiplatelet effect. Bleeding may occur 2–3h after protamine administration due to heparin rebound (heparin is resorbed from plasma proteins after protamine has been cleared from the circulation) which is treated by additional protamine administration. The **Horrow classification** lists three main types of protamine reaction.

Horrow classification of protamine reactions

- I: hypotension from rapid administration.
- II: anaphylactic reactions.
- III: catastrophic pulmonary vasoconstriction.

Type II reactions are subdivided into IIa true anaphylaxis, IIb immediate anaphylactoid, and IIc delayed anaphylactoid reactions.

Type I reaction

Protamine induces hypotension independently of the heparin–protamine complex, mediated by histamine release. This is dependent on the rate of administration. The fact that some patients become much more hypotensive than others probably reflects reduced myocardial reserve. Give heparin slowly (over 5min). Treat hypotension with careful volume transfusion from the pump. Inotropic support may be necessary.

Type II reactions

True anaphylaxis to protamine (Horrow type IIa) is mediated by a specific antiprotamine IgE antibody, and does not require the heparin–protamine complex. It requires previous exposure to protamine and is commoner in diabetics taking protamine-zinc insulin, patients with prior protamine exposure, and possibly patients with fish allergies. Vasectomized men are theoretically at higher risk because of cross-reactivity between antibodies to human sperm protamine and fish sperm protamine. Massive release of histamine and other inflammatory mediators (leukotrienes, kinins, prostaglandins) leads to acute bronchospasm, hypotension as a result of coronary vasospasm and systemic vasodilation, and angioedema. Respiratory and circulatory collapse can ensue. Anaphylactoid reactions are similar but less dramatic, and probably mediated by the complement pathway. Management for anaphylactic collapse is:

- Discontinue protamine immediately.
- 10mL of 1:100,000 adrenaline iv.
- 100mg hydrocortisone iv.
- 10mg chlorphenamine (Piriton®) iv.
- 100% O_2 if necessary, with nebulized bronchodilators.

Type III reaction

Acute pulmonary vasoconstriction leading to RV failure, circulatory collapse, and sometimes bronchospasm is caused by activation of the complement pathway and massive release of thomboxane A2 (a potent vasoconstrictor) in the lungs, after even minute doses of protamine, in <1% of patients. Acute RV failure due to poor myocardial protection causes a similar picture, with similar timing, and is managed the same way:

- Discontinue protamine infusion and give full dose of heparin.
- Give 10mL of 1:100,000 epinephrine.
- Start NTG and consider milrinone to reduce pulmonary hypertension.
- The initial episode usually may resolve, but if the heart is too compromised to weather this period recannulate (with internal massage if needed) and go back on bypass as soon as safe to do so.
- Slow administration of protamine is usually well tolerated a second time.

RV or LV failure

Failure of the heart to wean from bypass is manifested by low systemic pressures, high filling pressures, PA that feels very firm to touch, and a distended and dyskinetic ventricle. Myocardial stunning (📖 p74) accounts for an acute deterioration in LV function. Repair of severe mitral regurgitation may unmask poor LV function by removing the low-resistance regurgitant pathway (📖 p401). Incomplete revascularization or inadequate myocardial protection results in reduced LV and/or RV function.

- If you have already started protamine, stop and give full-dose heparin.
- Go back on bypass, ensure the heart is fully decompressed (vent PA).
- Some surgeons recommend a period on bypass equal to 1/3 of the ischaemic (cross-clamp) time to allow stunned myocardium to recover.
- Escalate inotropic support (📖 p198).
- If necessary, pressure in the aortic root, pulmonary artery and LA can be measured directly using a manometry line handed out to the anesthesiologist and a 23G (blue) needle, to help assess preload and afterload, and confirm invasive monitoring.
- If the heart fails to wean with appropriate support, insert an IABP.

Pump failure

Pump failure is unusual. Causes include electrical failure (rare because most operating rooms have back-up generators, and many CPB consoles have their own battery packs), and oxygenator failure (0.25 per 1000 pump runs), which is associated with the inadvertent administration of protamine during CPB that causes massive clot throughout the bypass circuit and oxygenator. In the event of total electrical failure the bypass circuit can be hand cranked. Oxygenator failure (characterized by inadequate oxygenation despite increasing FiO_2, and high transmembrane pressures) requires circulatory arrest (maximum of 5min safe at 36°C, longer at lower temperatures 📖 p125) to replace the oxygenator and recirculate: a drill that all perfusionists are taught and rehearse regularly. *Inadvertent administration of protamine requires a new pump circuit as well as administration of heparin. To avoid this potentially lethal error protamine is never drawn up until the instruction to give it is made.*

Emergency bypass scenarios 2

Aortic dissection

Aortic dissection from aortic cannulation has an incidence of about 0.05%. It can occur at the clamp or antegrade cardioplegia, but more commonly at aortic cannulation where it is recognized by:

- Purple hematoma with bleeding at a distance from the cannula site.
- Blood does not rise up the aortic cannula on cannulation.
- On removing the tubing clamp, blood does not flow normally.
- The patient becomes profoundly hypotensive with low RAP.
- Ischemic changes develop on the EKG.
- 'No swing' on connecting up the arterial line with the aortic cannula.
- Low systemic pressure confirmed by the Teicos and palpation.
- High line pressures on infusing prime via the aortic cannula.
- Poor venous return.
- *TEE or epiaortic ultrasound shows dissection flap.*

► If bypass has to be discontinued for any reason always clamp the venous line, or the patient will exsanguinate into the venous reservoir.

The patient needs emergency repair of the dissection, usually with an interposition graft, in addition to the planned surgery. Stop bypass immediately. Clamp the venous line. Place the aortic cannula into the RA so volume can be given rapidly as required, and proceed to femoral-caval bypass immediately, or occasionally cannulation of distal arch. Survival of aortic dissection recognized intraoperatively is about 85%.

IVC laceration

The IVC may be lacerated by passage of venous cannula, attempts at snaring the cavae, or dissection during redo surgery. Intrapericardial tears are obvious from venous hemorrhage into the pericardial sling, classically after termination of bypass. Intra-abdominal tears result in:

- Poor venous return.
- Low systemic and filling pressures.
- Loss of volume from the bypass circuit requiring volume replacement.

Perform a patch repair with autologous pericardium or saphenous vein graft as the tissue is usually too friable to close primarily, and to avoid stenosis. Endovascular stents have been used for intra-abdominal tears.

Airlock

If substantial amounts of air enter the venous lines (from perforation of the RA or cavae, dislodged venous cannula, or communication between an open LA and a cannula in the RA) the siphon may be interrupted, resulting in cessation of venous return, emptying of the venous reservoir and interruption of bypass if not addressed quickly. Reposition the cannula if grossly displaced. Chase the air down to the venous reservoir by elevating each section of the venous line from heart to pump in turn. It may be necessary to refill the venous line from the table. Vacuum-assisted venous drainage almost eliminates this problem: airlocks only occur with substantial air during periods of low flow.

Air embolism

In contrast to venous air, arterial air can be catastrophic. Air can enter the heart from the operative field, the pump, or anesthetic lines. Fatal air embolism can occur from amounts as small as 0.25mL/kg entering the left side of the heart and the coronary and cerebral circulation. Although right-sided air embolism is better tolerated in the absence of PFO, it can lead to severe pulmonary hypertension. Surgical sources of air include aortic cannulation, cardiotomy for valve surgery, entraining air with aggressive RSPV venting, entraining air from opened LAD when venting the aortic root. Air tends to collect in the LA, in the right superior pulmonary vein, and within the trabeculae of the LV, which explains the rationale for standard de-airing techniques (📖 p114). Air embolism from the CPB circuit can occur if the CPB reservoir empties: maintaining volume of the reservoir by interrupting bypass if necessary is a fundamental safety principle. Accidental disconnects, punctures, taps open to air, oxygenator or vacuum-assist malfunction can all potentially result in air embolism. The treatment of air embolism depends on the severity and timing:

- Coronary air embolism is best managed by continuing CPB, if appropriate pinching the aorta distal to the aortic cannula to increase coronary perfusion pressures for a few minutes.
- Hypothermia, retrograde coronary and cerebral perfusion (see Box 4.1) and de-airing procedures (📖 p114) are useful adjuncts.
- Ventilating the patient with 100% oxygen and decreasing cerebral oxygen demand by packing the patient's head in ice may help.
- If air embolism occurs before or after CPB with cardiac arrest, place the patient head down in the left lateral position and institute CPR.
- Drug therapy is first aimed at increasing BP to 'flush' air through to the venous circulation, and then limit tissue damage with steroids, diuretics, antiplatelet agents, anticonvulsants, and barbiturates.
- The surgical procedure should be completed regardless of the timing.
- Compression in a hyperbaric chamber and ventilation with 100% oxygen is the definitive management, but logistics make this difficult.

Box 4.1 Treatment of massive air embolism on bypass

- Stop bypass immediately
- Clamp venous and arterial lines, aspirate air, start cardiac massage.
- Put patient head down, occlude carotids and place ice around head.
- Remove aortic cannula from ascending aorta.
- Connect venous and arterial lines together so that air can be purged from the bypass circuit by the perfusionist, or fill from table.
- If large quantities of air have entered the cerebral circulation insert the aortic cannula into the SVC and snare it.
- Begin retrograde cerebral perfusion at 20°C at 1–2L/min for 1–3min, until no more froth is seen entering the aorta.
- Otherwise reconnect the lines to the appropriate cannula and resume bypass with active aortic root venting.
- Give 2–4g methylprednisolone and 25g mannitol.

Alternative approaches

Alternatives to ascending aortic cannulation

There are several alternative arterial cannulation sites if the ascending aorta is unsuitable for any of the following reasons:

- Porcelain aorta (a completely calcified aorta).
- Severe atherosclerosis with or without liquid intramural disease.
- Surgery involving the distal ascending aorta or aortic arch.
- High-risk re-do surgery (📖 p344).
- Small incision surgery (📖 p458).

Elsewhere on the aorta or innominate artery

- Underside of the distal aortic arch is most likely to be disease free.

Femoral artery: left or right.

- Useful in emergency, re-ops, and minimally invasive approaches.
- Vasculopathy relative contraindication.
- Disadvantages: increased risk of stroke with retrograde arterial perfusion in patients with atherosclerotic aortic disease. Risk of peripheral limb ischemia almost eliminated by cannulating via an 8mm Dacron side graft sewn end-to-side onto the artery with 5/0 Prolene.
- *Technique*: 3–4cm incision in skin crease over femoral pulse, blunt dissection down onto femoral artery (may be very adhesed if prior arterial punctures), give 5000U of heparin. Either modified Seldinger insertion of 22F FemFlex cannula or sidegraft (see axillary). Arteriotomy should be repaired at end of case.

Axillary artery: usually right

- Particularly useful for selective perfusion in arch surgery (📖 pp125, 449), but also good for same indications as femoral cannulation.
- Less likely to be involved in vasculopathy, and antegrade arterial perfusion means stroke risk greatly reduced compared to femoral.
- Disadvantage is proximity to brachial plexus and obvious incision.
- Lower risk of peripheral ischemia as more collaterals than femoral.
- *Technique*: 4–5cm oblique incision 1cm below lateral 1/3 of clavicle, divide pectoralis major with cautery down to pectoralis minor, divide pec minor for easiest exposure, axillary artery lies immediately beneath pec minor. Give 5000U of heparin. Silastics to control 2–3cm of artery avoiding brachial plexus fibres, and ligate all arterial branches (if you use silastic ties these can be removed at end of case). Apply vascular clamps proximally and distally, longitudinal arteriotomy, sew 8mm Dacron sidegraft with continuous 6.0 Prolene, and cannulate with 24F Sarns (remove cuff). Multiple silk ties around graft and cannula. Clamp cannula, remove vascular clamps from axillary artery, de-air and connect in usual fashion. Use distal axillary sling to prevent hyperperfusion of arm. Staple or oversew graft at end of case.

Alternatives to caval cannulation

If it is not ideal to cannulate centrally (minimally invasive, or re-operative surgery), the femoral or axillary vein can be cannulated after heparinization, percutaneously or under direct vision, passing a long perforated venous cannula

carefully up into the RA under TEE guidance using Seldinger technique. If done open, place a 5/0 purse-string for easy closure at the end of the case.

- Fem-fem bypass is bypass via femoral artery and vein.
- Fem-caval bypass is via the femoral artery and RA.
- Aorto-femoral bypass is via the aorta and femoral vein.
- Axillo-caval bypass is via the axillary artery and RA.

Alternatives to cross-clamping the ascending aorta

- If the aorta cannot be clamped (porcelain aorta, severe atherosclerosis, imminent rupture) there are several options:
- **Beating heart on bypass**, where the aorta is not clamped and the coronaries are perfused with blood from the aortic cannula for the duration of surgery can be used to perform coronary mitral, tricuspid, or pulmonary valve surgery. If there is more than minimal aortic insufficiency it can be quite difficult to get decent visualization of the mitral valve because of blood flooding the field.
- **Fibrillating heart on bypass** can be used in conjunction with beating heart or on its own. If the aortic valve is incompetent the LV must be adequately vented as soon as the heart stops beating to prevent distension. Fibrillating the unclamped heart on bypass may occur as part of a deliberate hypothermic protection strategy (usually < 32°C), or can be done using a fibrillator at normothermia, in order to minimize the risk of air being ejected from the LV during mitral surgery performed without a cross-clamp.
- Neither of these are options for aortic valve or aneurysm surgery. In this case **deep hypothermic circulatory arrest** (DHCA) (📖 pp124–125) must be instituted if it is impossible to clamp the aorta, via any of the cannulation strategies outlined opposite. It is possible to resume cardiopulmonary bypass once a graft has been sewn onto the transected distal aorta, de-aired and clamped. (📖 p125, 📖 p449).
- **Off-pump coronary surgery** (📖 pp304–305, 335).

Persistent left superior venal cava

A left SVC is present in 0.3–0.5% of the general population, and in 2–10% of patients with congenital heart disease. It drains the left subclavian vein and jugular veins into the RA via the right coronary sinus in 90% of cases, and into the LA in 10% of cases. Suspect the presence of a LSVC if the innominate vein is small or absent; TEE may show an enormous coronary sinus.

- If the right heart is not opened and a two-stage cannula is used, a LSVC poses no problems. Retrograde cardioplegia will not work.
- If bicaval cannulation is used, the problem of blood flow into the RA causing right heart distension, or occluding the operative field can be solved by snaring the LSVC, unless there is absence of the RCVC or innominate vein in which case cerebral edema would result.
- If the LSVC cannot be snared, it can be cannulated via the coronary sinus, vented or cannulated distally with a caval cannula.
- Induction of deep hypothermic circulatory arrest with a single venous cannula is used in infants.

Myocardial protection

Key facts

- Myocardial protection should minimize myocardial ischemia and damage during bypass, primarily by reducing myocardial O_2 consumption.
- This is usually achieved by rapid diastolic arrest, mild or moderate hypothermia, buffering, avoiding substrate depletion, and intracellular edema using cardioplegia, but there are alternative options (Box 4.2).

Cardioplegia

Rapid diastolic arrest reduces MVO_2 by 90% and minimizes depletion of high-energy phosphates. Reducing the arrested heart from 37°C to 32°C drops MVO_2 by 90%: lower achieves little extra benefit. Most strategies rely on concentrated K^+: St. Thomas' and Bretschneider are crystalloid solutions. Blood cardioplegia, e.g., Buckberg, is used for its intrinsic O_2 carrying and buffering capacity: it may better protect in poor LV or long ischemic times. Crystalloid plegia is given from a pressurized bag. Blood plegia is given from the pump enabling accurate control and monitoring of volume, pressure, flow and temperature, and is made up from the patient's blood once CPB is commenced.

Antegrade cardioplegia

Root cardioplegia

Plegia is given into the aortic root via a simple trocar, e.g., Medicut, or via a cannula with a side arm root vent, e.g., Stockert, which needs a partial-thickness 4/0 Prolene purse-string 2cm from the sinotubular junction.

- Flush the cardioplegia line so that it is completely free of air.
- Insert the cannula taking care not to damage the back wall of the aorta.

Box 4.2 Types of myocardial protection

- Myocardial protection may or may not involve cardioplegia.
- Non-cardioplegic methods include 'cross-clamp fib' (📖 p325), fibrillating (📖 p121) or beating heart (📖 p325) on bypass, and DHCA (📖 pp124–125).
- Cardioplegia may be given intermittently or continuously, given antegradely or retrogradely, be crystalloid (which can be intracellular, e.g., Bretschneider or extracellular, e.g., St. Thomas'), or blood (warm or cold) and high (20mmol/L) or low (10mmol/L) K^+.

Protection in high-risk cases

In cases with long ischemic times or severe RV and LV dysfunction protect the heart by maintaining a uniform cold temperature using systemic cooling, topical hypothermia, insulating pads (to stop liver warming base of heart), venting to prevent collateral flow warming heart, total CPB (📖 p103), myocardial temperature monitoring, frequent (15min) doses of cardioplegia, running cold blood the rest of the time. Retrograde cardioplegia facilitates this. A hot shot is given (📖 p123).

- Snare to the purse-string, remove the trocar, and clamp the cannula.
- Ask the perfusionist to trickle the plegia and connect the cannula.
- Ask the perfusionist to reduce flows while you cross-clamp the aorta between the aortic cannula and the antegrade cardioplegia cannula.
- Ask the perfusionist to give cardioplegia—either until complete electromechanical arrest is achieved, or a standard volume (800mL–1L).
- Check the LV and RV intermittently to ensure they are not distending and are cooling uniformly, and that the aortic root is pressurized.
- Once all cardioplegia is given and the heart is arrested, clamp or remove the cardioplegia cannula and empty the LV by venting the aortic root and/or gently squeezing the ventricles.

Ostial cardioplegia

This technique requires an aortotomy. The aorta is opened after clamping, and pump sucker placed in the LV, and the ostia visualized. A balloon-tipped cannula is attached to the cardioplegia line: size 6–7 for the left main, size 5–6 for the smaller RCA. If plegia does not run well, adjust the angle, or try a smaller cannula. Beware of inadvertently selectively cannulating the circumflex or LAD in the patient with a short LMS.

Retrograde cardioplegia

Insertion is described on 📖 p111. Retrograde cardioplegia does not protect the RV if the cannula is inserted too far, bypassing the great cardiac vein. It is a useful adjunct when high-grade coronary stenoses make antegrade delivery inefficient, or there is a need to give continual cardioplegia during long surgery (e.g., root replacement, complex mitral repair) without adjusting the heart or stopping the flow of surgery. Venovenous channels mean up to 60% drains back into the heart via the Thebesian veins. Satisfactory administration can be confirmed by: (1) decrease in myocardial temperature; (2) sustained increase in coronary sinus pressure from 0–10mmHg to 50mmHg while giving 100mL/min; (3) filling the cardiac veins over the RV; (4) dark blood issuing from both coronary ostia.

Terminal warm blood or 'hot shot'

Before unclamping (and usually while performing proximal anastomoses or closing heart, warm blood (no extra K^+) is usually or retrograde which may restore energy stores and may improve ventricular recovery (although data is very mixed), as well as help with de-airing.

Failure of the heart to arrest

- High-grade proximal RCA lesion (no plegia reaching sinus node).
- Blood cardioplegia is slower to arrest than crystalloid, and retrograde arrest is slower than antegrade.
- Cross-clamp not properly applied (plegia diluted with aortic blood—stop giving, reduce bypass flows, and reapply clamp more carefully).
- L/RV distension (vent, improve drainage, check clamp, caval snares).
- AI (plegia filling LV—pinch root, give ostial plegia or give retrograde).
- Cardioplegia incorrectly made up, or not cold (check temperature).

Cerebral and spinal protection

Key facts

- *Cerebral protection* is necessary when bypass cannot supply the head vessels, e.g., during surgery of the distal ascending aorta and arch.
- There are two modalities: hypothermia, which is by far the most important, and retrograde or antegrade selective cerebral perfusion.
- *Spinal cord protection* includes hypothermia and CSF drainage aimed at reducing risk of paraplegia in thoracoabdominal surgery (📖 p436).

Hypothermia

During coronary and valve surgery most surgeons cool actively or passively (drift) to about 32°C. At 37°C irreversible cerebral damage occurs after 2–3min of ischemia. Cerebral O_2 consumption is reduced by approximately 5% per 1°C drop between 37°C and 22°C. At 20°C cerebral O_2 consumption is 20% of normothermia. Although this suggests a safe circulatory arrest time of 15–20min, clinical evidence suggests that 30min is tolerated, rising to 45min at 18°C as other factors including pH and suppression of neuronal function may pay a role.

Acid–base changes with hypothermia

Normal pH is 7.40 and normal $PaCO_2$ is 5kPa in blood at 37°C. A sample of arterial blood in an airtight syringe has a PCO_2 that varies directly with temperature: decreasing the temperature of a solution increases the solubility of a given gas, decreasing its partial pressure. pH, on the other hand, varies inversely with temperature: the greater the CO_2 content in solution, the more H^+ in solution. CPB is not an airtight system: CO_2 can be added or taken away. So as hypothermia is induced the perfusionist has the choice of leaving total blood CO_2 constant, and letting pH and $PaCO_2$ find their own levels dictated by the temperature, or adding CO_2 to maintain $PaCO_2$ at 5kPa and maintain pH at 7.40. The former is known as *alpha-stat* CPB, and the latter is known as *pH-stat* CPB (Box 4.3). pH-stat management seems logical as most enzymes depend on optimal pH for their function, but the depression of metabolism that occurs with increasing acidosis characterized by alpha-stat management may be beneficial in terms of cerebral and myocardial protection. Recent evidence suggests that alpha-stat management gives better results in moderate hypothermic CPB. During deep hypothermic CPB best results may be achieved with a mixed strategy: pH stat during cooling, alpha-stat during rewarming, reperfusion and termination of bypass.

Deep hypothermic circulatory arrest

After institution of bypass the patient is cooled to a point where bypass can be discontinued. This technique is used in distal ascending aorta and

Box 4.3 Alpha-stat and pH-stat

- *PH-stat tries to keep the pH stable* by adding CO_2 as temperatures fall and removing it as they rise.
- *Alpha-stat does not add anything*. $PaCO_2$ rises and pH falls with falling temperature.

arch repair (📖 p448), and sometimes during pulmonary embolectomy and stage II elephant trunk. Due to decreased tissue metabolism and O_2 demand, even a small degree of systemic hypothermia allows lower pump flows, less blood trauma, better myocardial protection, and better organ protection than normothermic perfusion. Hypothermia decreases blood flow to vascular beds in proportion with reduced metabolic demands, most pronounced in skeletal muscle and the extremities.

Left heart bypass

This is a useful adjunct in surgery on the descending aorta, where the heart does not need to be arrested, but the distal descending aorta and lower limbs need perfusion during the period of descending aorta cross-clamp. The LA is cannulated and oxygenated blood is siphoned into a reservoir, and pumped into a femoral line. An oxygenator is not required as blood is oxygenated by the lungs. A heat exchanger is optional as the patient is allowed to cool passively. Without these components there is less need for heparinization: 100U/kg heparin is normally given, and the ACT is maintained at <200s. Flows are maintained at 75% of full flow and MAPs distal to the clamp at 55–70mmHg.

Safe duration of circulatory arrest

- 36°C < 2–3min.
- 32°C 3–5min.
- 28°C 10min (moderate hypothermia).
- 20°C 30min (DHCA).
- 15°C 45min (DHCA).

Cerebral perfusion

Retrograde cerebral perfusion

Bicaval cannulas and a venous to arterial bridge are required: when retrograde perfusion is required bypass is discontinued, the IVC cannula is clamped, the SVC cannula is snared, and the perfusionist uses the bridge to pump oxygenated blood up the venous line. Deoxygenated blood is drained from the head and neck vessels. Low, near-continuous flow is used by some surgeons during hypothermic circulatory arrest. The common observation of dark, deoxygenated blood flowing out of the neck vessels suggests that cerebral oxygen utilization occurs.

Antegrade (selective) cerebral perfusion

This is used in arch surgery to reduce DHCA time. The brachiocephalic artery and left common carotid artery can be directly cannulated with a retrograde cardioplegia cannula. Or, if the axillary artery has been cannulated and the head and neck vessels anastomosed to a trifurcation graft the head can be selectively perfused from the axillary artery (📖 p449). This reduces the risk of embolization from instrumenting the vessels, and keeps the field clearer. Flows are maintained at 10mL/kg, MAPs 40–50mmHg, and temperature around 20°C.

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